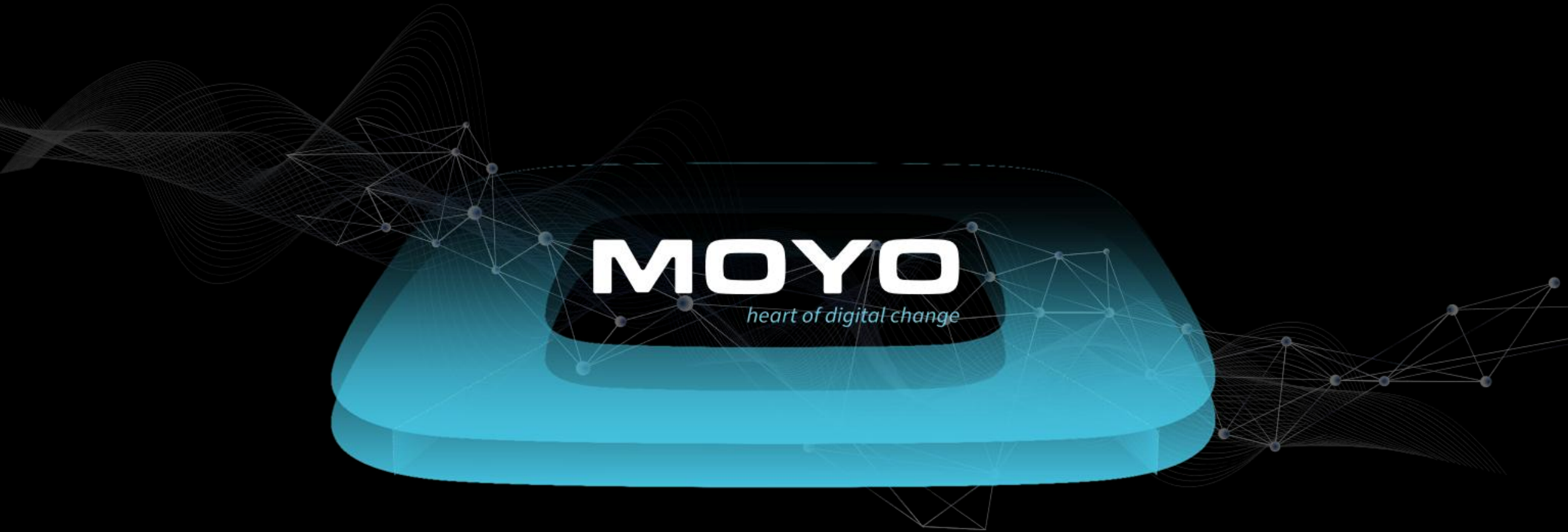
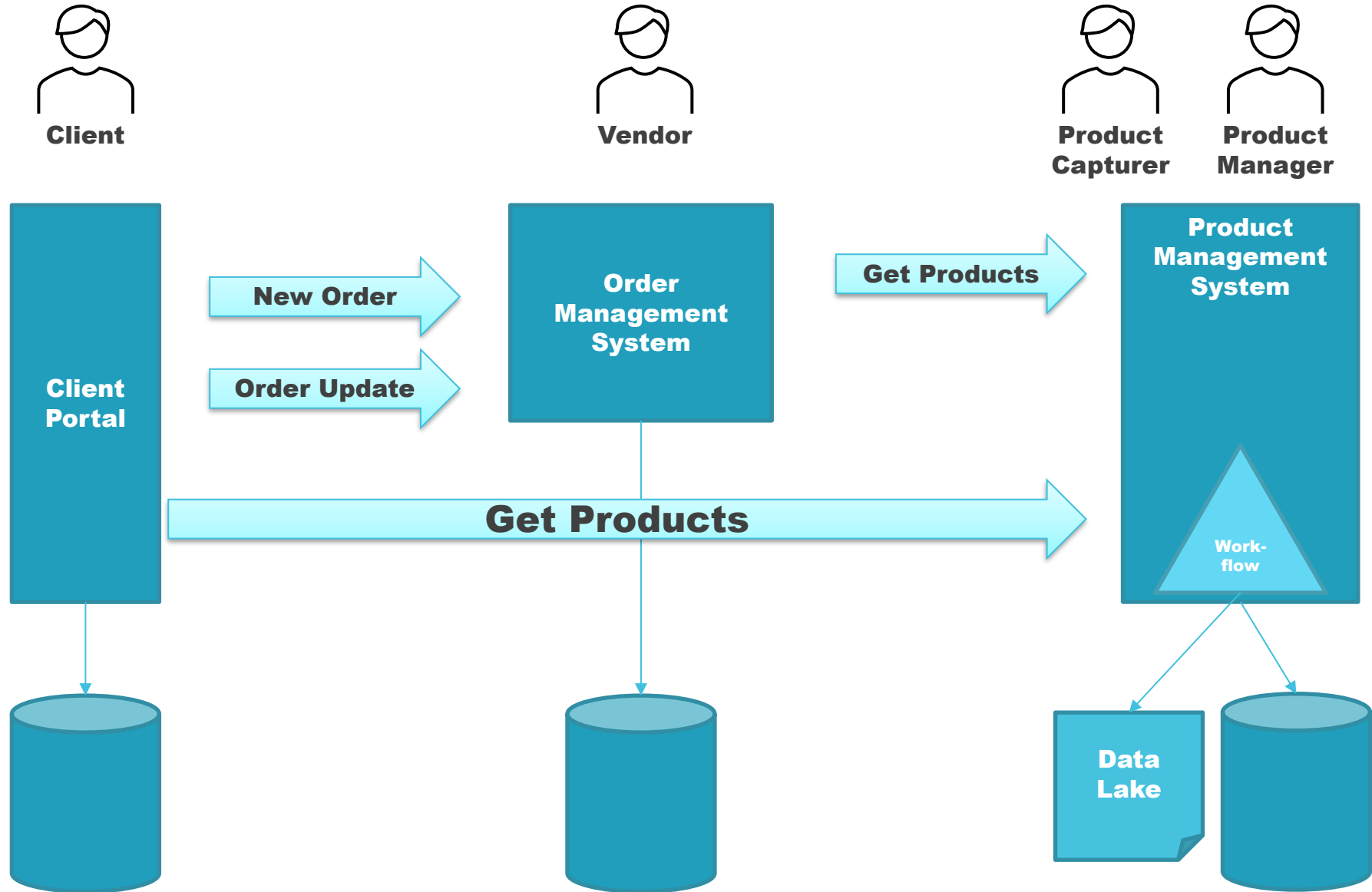




Online Order Solution – Case Study v1.3

Software Development Solutions

ONLINE ORDER SOLUTION.



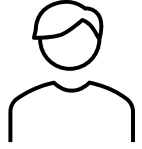
IMPLEMENTATION PLAN.

- The candidate has **the following options:**
 - The candidate can decide which of the 3 systems of the overall solution they want to implement
 - ❖ Client Portal
 - ❖ Order Management System
 - ❖ Product Management System
 - Technology focus
 - ❖ Microsoft focused Stream (with Modern UI technology options)
 - ❖ Documentation
 - ❖ Provide Detailed Solution Architecture of the entire Solution
 - ❖ Provide Detailed ERD of the entire Solution



MICROSOFT FOCUSED STREAM

CLIENT PORTAL.



Client



This is a system that allows:

- A client to login
- Browsing of Products (basic: 1 table)
- Browsing & viewing orders
- Adding of new Orders

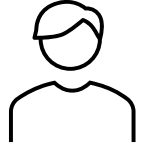
Technology:

- Angular Front End
- OAuth 2.0/OpenID Connect
- C# (.Net Core)
- Entity Framework (DB First)
- Asynchronous Communication to Order Management Platform (new Orders)
- Asynchronous Communication from Order Management Platform (Status updates)
- Synchronous Communication with Product System
- Public Cloud based PaaS



Microsoft
Focused Stream

ORDER MANAGEMENT SYSTEM.



Vendor



Microsoft
Focused Stream

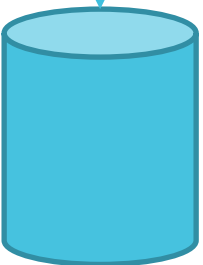
**Order
Management
System**

This is a system that allows:

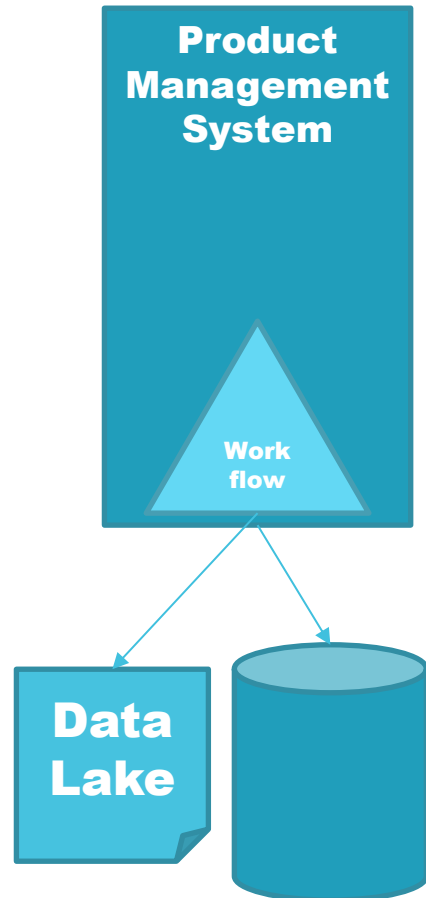
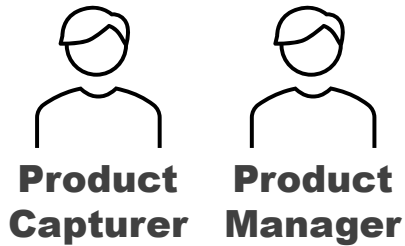
- A Vendor to login
- A Vendor to configure prices per Product
- A Vendor to configure stock per Product (inventory)
- The system to receive asynchronous Events regarding Orders
- The system to allocate Orders based on price & inventory to specific Vendors

Technology:

- Angular Front End
- OAuth 2.0/OpenID Connect
- C# (.Net Core)
- Entity Framework (Code First)
- Asynchronous Communication from Client Portal (new Orders)
- Asynchronous Communication to Client Portal (Status updates)
- Synchronous Communication with Product Management System
- Public Cloud based PaaS



PRODUCT MANAGEMENT SYSTEM.



This is a system that allows:

- A Product Capturer & Manager to login
- CRUD functionality on Products
 - Create, Read, Update = Capturer
 - Approve (Update & SoftDelete) = Manager
- Workflow functionality to approve changes to Products (Manager only)

Technology:

- Angular Front End
- OAuth 2.0/OpenID Connect
- Authorisation functionality important
- C# (.Net Core)
- Entity Framework (Code First)
- Data Lake for Approved content (quick retrieval)
- Synchronous Communication from Client Portal
- Synchronous Communication from Order Management System
- Public Cloud based PaaS



GENERAL

SOLUTION PRINCIPLES AND NOTES.

- Less is more
- Software Architecture considerations are critical
- Assume multiple subsequent enhancements would be required, so a good foundation is key.
- Assume large enterprise with large customer base.
- Public Cloud Platform (e.g. Azure)
- AI is allowed but it is CRITICAL that you understand what you (and the AI) are doing
- Paas > IaaS
- CI/CD beneficial
 - Infrastructure as Code not critical